

Code Layout, Readability & Reusability

Recommended repository organization

Date	30 September 2026
Team ID	SWTID-2026-9380
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Team	Lavinraj (Team Leader), Subash, Rahul, Hariharansudhan
Maximum Marks	4 Marks

Path / module	Responsibility
app/ui/	Profile form, generation status, plan summary, session details and export view.
app/domain/	Typed profile and plan models, business rules and progression logic.
app/services/gemini	Model adapter, configuration, request timeout and provider error mapping.
app/services/prompt	System instruction and profile-to-prompt formatting.
app/services/validation	Input schema, generated-plan schema and safety checks.
app/api/	Authenticated server endpoints for generation and history.
tests/unit/	Validation, prompt, parsing and progression tests.
tests/e2e/	Critical user journeys and accessibility checks.
docs/	Architecture, setup, API contract, test evidence and demo script.

Quality conventions

- Use descriptive names and small single-purpose functions.
- Represent user profiles and plans with typed schemas.
- Centralize model configuration and error mapping.
- Keep prompts versioned and independently testable.
- Use reusable exercise, day-plan and feedback components.
- Record decisions and limitations; never commit environment secrets.